1. **Aim/Overview of the practical**: Write a Program based on K-means clustering algorithm.

   (b) Write a program for multiple and polynomial regression.

2. **Objective of the practical:** To implement machine learning algorithms including K-means clustering, multiple regression, and polynomial regression for data analysis and prediction.

3. **Code for experiment/practical:**

   Multiple Regresssion

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

# Load data
df = pd.read_csv("Housing.csv")

# Select numerical features and target
X = df[["area", "bedrooms", "bathrooms", "stories", "parking"]]
y = df["price"]

# Scale the features
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2,
random_state=42)

# Train the model
model = LinearRegression()
model.fit(X_train, y_train)

predictions = model.predict(X_test)

# Evaluate
mse = mean_squared_error(y_test, predictions)
r2 = r2_score(y_test, predictions)

print("Mean Squared Error (MSE):", mse)
print("R² Score:", r2)
print("Predicted prices:", predictions)

# Plot actual vs predicted
plt.figure(figsize=(8,5))
plt.scatter(y_test, predictions, color='blue', label='Predicted vs Actual')
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--', label='Perfect
Prediction Line')
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted House Prices")
plt.legend()
plt.grid(True)
plt.show()
```

Polynomial

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.metrics import r2_score
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures

#  Load the dataset
df = pd.read_csv("Ice_cream selling data.csv")
df.columns = ["Temperature", "Sales"]

#  Input the data
X = df["Temperature"].values.reshape(-1, 1)
y = df["Sales"].values

# Transform to Polynomial Features (degree 2)
poly = PolynomialFeatures(degree=2) #quadratic regression
X_poly = poly.fit_transform(X)

#  Fit the Model
model = LinearRegression()
model.fit(X_poly, y)

# Print the Coefficients
print("Intercept (β0):", model.intercept_)
print("Coefficients (β1, β2):", model.coef_[1:])

#  Make Predictions
X_test = np.linspace(X.min(), X.max(), 100).reshape(-1, 1)
X_test_poly = poly.transform(X_test)
y_pred = model.predict(X_test_poly)

print("R² Score:", r2_score(y, model.predict(X_poly)))

# Plot the Results
```

```
plt.scatter(X, y, color='red', label='Original Data')
plt.plot(X_test, y_pred, color='blue', label='Polynomial Fit (degree 2)')
plt.xlabel('Temperature (°C)')
plt.ylabel('Ice Cream Sales (units)')
plt.title('Polynomial Regression (Degree 2) - Ice Cream Sales')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

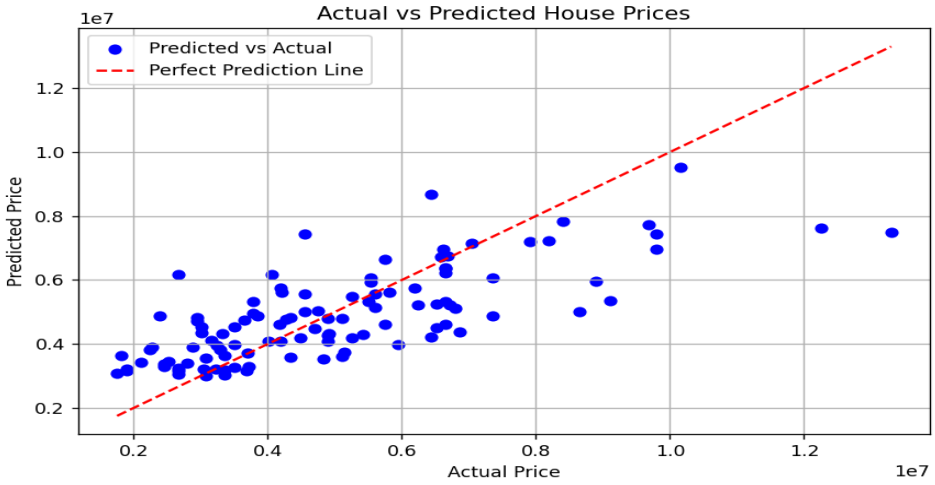
K-means Clustering

```
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt
import seaborn as sns

data = pd.read_csv("heart.csv")

#  features for clustering
X = data[['age', 'chol', 'trestbps', 'thalach']]

# Scale the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Apply KMeans
kmeans = KMeans(n_clusters=3, random_state=42)
data['Risk_Cluster'] = kmeans.fit_predict(X_scaled)

# Print cluster centers (scaled values)
print("\nCluster Centers (scaled):")
print(kmeans.cluster_centers_)

plt.figure(figsize=(8,6))
sns.scatterplot(data=data, x='age', y='chol', hue='Risk_Cluster', palette='viridis', s=80)
plt.title("K-Means Clustering: Age vs Cholesterol (Risk Groups)")
plt.xlabel("Age")
plt.ylabel("Cholesterol")
```

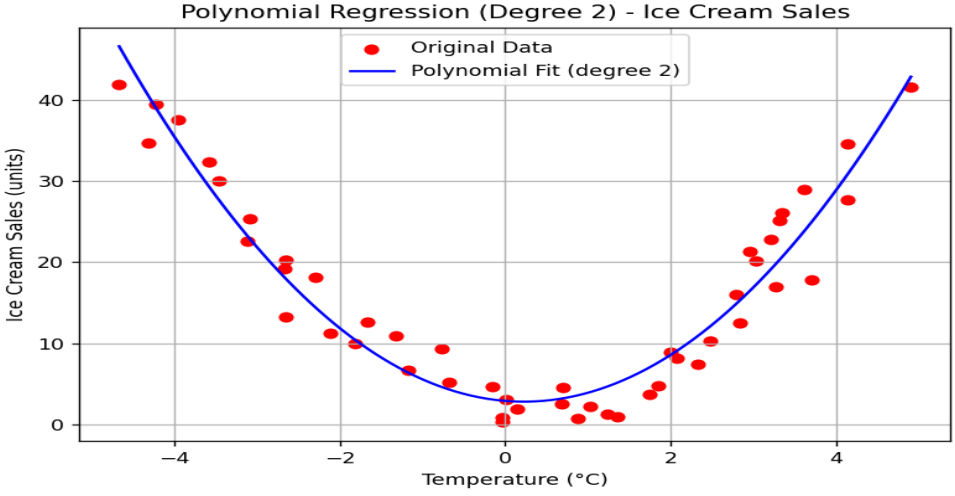
```
plt.legend(title="Risk Cluster")
plt.show()

plt.figure(figsize=(8,6))
sns.scatterplot(data=data, x='age', y='thalach', hue='Risk_Cluster', palette='plasma', s=80)
plt.title("K-Means Clustering: Age vs Max Heart Rate (Risk Groups)")
plt.xlabel("Age")
plt.ylabel("Max Heart Rate")
plt.legend(title="Risk Cluster")
plt.show()
```

4. Output:

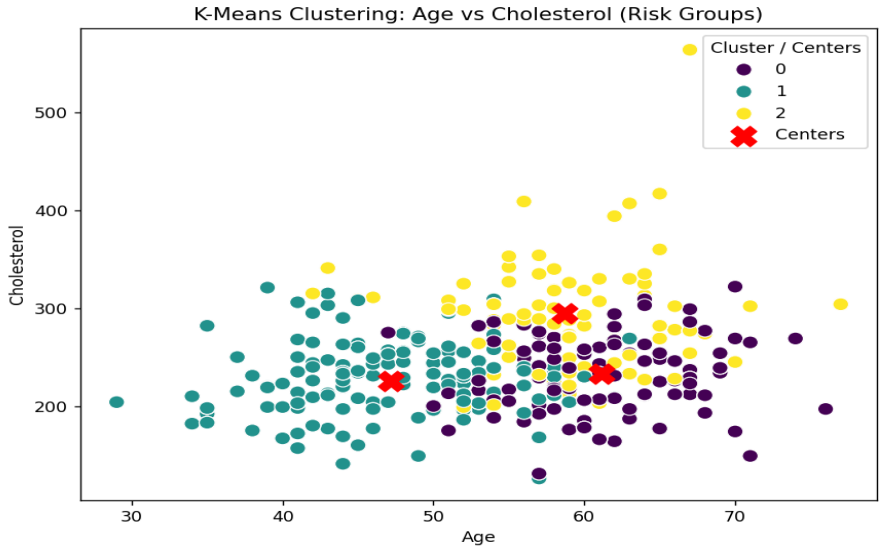
Multiple Regression

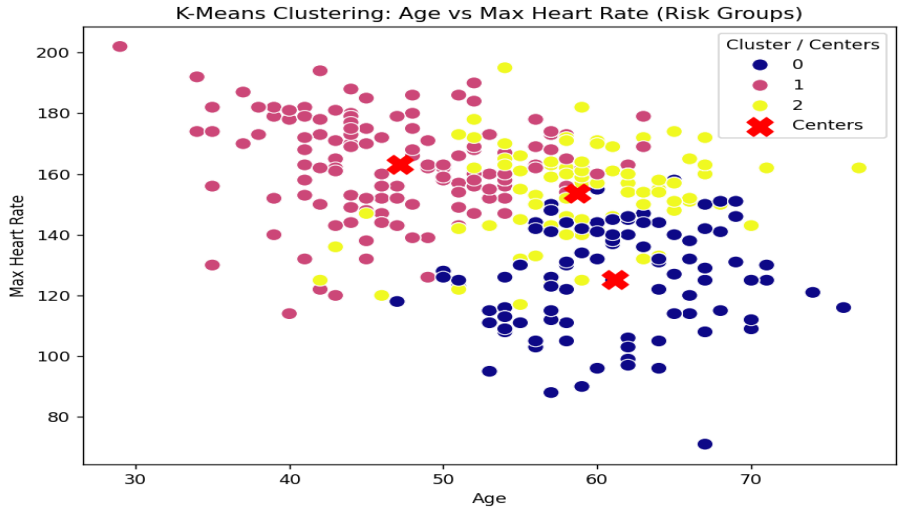


Polynomial



K-means Clustering





5. Learning Outcomes (What I have learnt):

- Learners will understand the concepts of unsupervised learning (K-means) and supervised learning (regression).
- They will gain skills to write Python programs that perform clustering and predictive modeling.
- They will analyze and interpret model outputs to identify patterns and make accurate predictions.